Fiscal Year:	FY 2004	**Task Last Updated:**	FY 03/17/2006
PI Name:	Banish, R. M. Ph.D.		
Project Title:	Thermophysical Property Measurements of Te-Based II-VI Semiconductor Compounds		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Electronic materials		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	banishm@email.uah.edu	**Fax:**	FY 256-824-6944
PI Organization Type:	UNIVERSITY	**Phone:**	256-824-6969
Organization Name:	University of Alabama Huntsville		
PI Address 1:	Center for Microgravity and Materials Research		
PI Address 2:	Mail Stop VBRH-D11, 301 Sparkman Drive		
PI Web Page:	http://www.cmmr.uah.edu/		
City:	Huntsville	**State:**	AL
Zip Code:	35899	**Congressional District:**	5
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	96-HEDS-02
Start Date:	02/01/1998	**End Date:**	04/30/2004
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	1	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Banks, Curtis	**Contact Phone:**	256 544-4437
Contact Email:	curtis.e.banks@nasa.gov		
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Alexander, John I.D. (Case Western Reserve University)		
Grant/Contract No.:	NCC8-1476		
Performance Goal No.:			
Performance Goal Text:			

Toward the task objective we are pursuing the following scientific tasks:
- Measurement of binary- and dopant (impurity) diffusivities in liquid (Te-based) semiconductors, over a temperature range of approximately 100 °C above the melting point;

- Measurement of thermal expansion on melting, and the thermal and solutal expansion in melts;

- Measurement of liquid thermal conductivity and/or diffusivity;

Toward completing the above scientific tasks we are pursuing the following hardware development goals:

- Design and construction of hardware for semi-automated measurement of diffusivities at several temperatures per sample;

Task Description:	- Design and construction of hardware for semi-automated measurement of thermal diffusivity at several temperatures per sample. Mass diffusivities are determined from temporal records of evolving concentration profiles through measurement of radioactive tracer emission. An initially solid sample contains a radiotracer (of the same element/compound) at one end. After melting radiation escaping preferentially through collimation bores in an isothermal liner/radiation shield surrounding the sample is monitored via two, advantageously placed, radiation detectors. Diffusivities are determined from a novel algorithm which permits data deduction from any continuous concentration distribution. To account for thermal expansion of the sample two pairs of radiation detectors are employed. Hence, diffusivity data can be gathered over a range of temperatures in a single experiment. Thermal diffusivities will be determined via a transient heat pulse methodology. The temperatures will be measured using high precision, rapid response, sapphire blackbody emitters coupled to silica optical fibers. Traditional thermocouples cannot be used in the Te-based systems studied here because the melts attack (dissolve) the thermocouple materials themselves as well as any metal sheaths used for protection. In fact, sapphire blackbody emitters are capable of providing better temperature resolution than thermocouples and themselves do not require recalibration after use. In addition, they are less susceptible to poisoning due to contamination of the container or the melt itself. A finer temperature resolution and higher sensitivity is particularly important for thermal diffusivity measurements of II-VI materials due to their, typically, low diffusivities. Presently, we are evaluating different geometries in an effort to develop a simple numerical algorithm similar to the one employed for mass diffusivity determinations (see the paragraph above). This research aims at advancing the understanding the melt structure and impact of the melts structure on solidification via understanding the binary and impurity diffusion mechanisms in Te-based II-VI semiconducting compound melts (i.e., CdTe, ZnTe, and HgTe). This will be done through analysis of accurate diffusivity measurements. In addition the species diffusivity measurements, the thermal diffusivities of these materials will be determined. Thermal conductivity values for liquid semiconductors extrapolated from the Weidermann-Franz relation (via the electrical conductivity) are too high, indicating that this relation is invalid for these materials. Finally, we are determining the melting point expansion in support of this research. Tellurium based II-VI semiconductor devices are the basis for a multi-million dollar "high tech" industry. These materials have a strong technological advantage for infrared and X-ray detectors, imaging devices and optical components. However, their current poor production yields results in high cost and limited availability for these devices. The yield for passive devices is typically less than 25%, and less than 5% for active devices. Numerical modeling efforts of the transport processes during solidification have lead to some improvements in device yield. However, these efforts currently suffer from a lack of realistic thermophysical input parameters, and in particular their temperature dependence. For numerical modeling efforts to make a meaningful contribution to improving device yields accurate values of these parameters are necessary. The experimental determination of transport properties such as diffusivities and conductivities is often contaminated by buoyancy-driven convective effects. The accurate determination of transport properties requires convective flow velocities to be eliminated or significantly suppressed. This is generally not possible under terrestrial conditions. Earlier liquid (metal) diffusion studies conducted on spacecraft have demonstrated the gain in precision resulting from the drastic reduction of convection in low-gravity environments.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	No progress reported this period.
Bibliography Type:	Description: (Last Updated: 02/27/2004)